

Assignment 2

Task 1:

Create a custom column that calculates the profit margin for each product. Use the formula: $\text{ProfitMargin} = (\text{ProductPrice} - \text{ProductCost}) / \text{ProductPrice}$. Name this column "ProfitMargin". Add a conditional column to classify products as "HighMargin", "Medium Margin", or "Low Margin" based on their profit margin. Define the thresholds: High Margin (> 0.5), Medium Margin ($0.2 - 0.5$), Low Margin (< 0.2).

Sol:

- A new column named ProfitMargin was created.
- This column shows the relative profitability of each product as a decimal value (e.g., 0.43 means 43% profit margin).

DAX Formula: Profit = products[ProductCost]-products[ProductPrice]

- Products are now categorized as High Margin, Medium Margin, or Low Margin.
- This classification helps in segmenting products based on profitability for better pricing and marketing strategies.
- ProfitMargin can be used in reports to compare profitability across categories or over time.

Table with 5 columns: ProductStyle, ProductCost, ProductPrice, ProfitMargin, Thresholds.

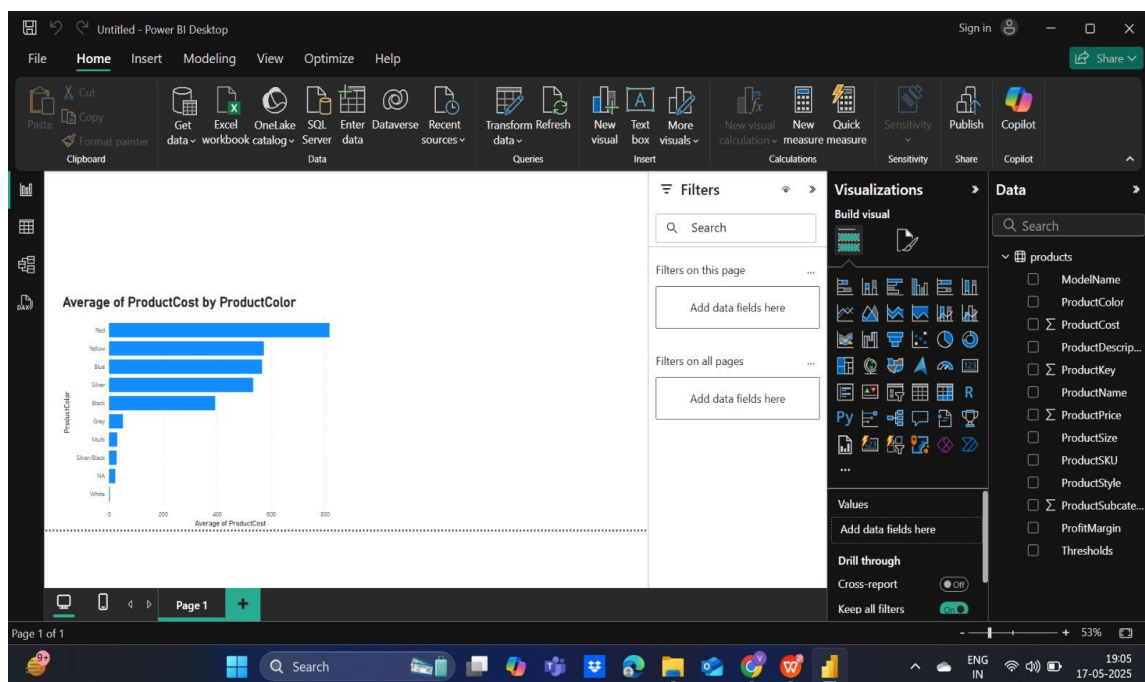
ProductStyle	ProductCost	ProductPrice	ProfitMargin	Thresholds
Valid	100%	Valid	100%	Valid
Error	0%	Error	0%	Error
Empty	0%	Empty	0%	Empty
1	0	13.0863	34.99	0.62598857 High
2	0	12.0278	33.6442	0.642500045 High
3	0	3.3963	9.5	0.642494737 High
4	0	3.3963	9.5	0.642494737 High
5	0	12.0278	33.6442	0.642500045 High
6	0	5.2052	8.6442	0.39996763 Medium
7	0	31.7244	48.0673	0.340000374 Medium
8	0	31.7244	48.0673	0.340000374 Medium
9	0	31.7244	48.0673	0.340000374 Medium
10	0	31.7244	48.0673	0.340000374 Medium
11	0	747.9682	1263.4598	0.408000001 Medium
12	0	747.9682	1263.4598	0.408000001 Medium
13	0	747.9682	1263.4598	0.408000001 Medium
14	0	747.9682	1263.4598	0.408000001 Medium
15	0	747.9682	1263.4598	0.408000001 Medium
16	0	176.1997	297.6346	0.407999944 Medium
17	0			

Task 2:

Use the Q&A feature to find out "What is the average product cost by product color?" and display the results as a bar chart.

Sol:

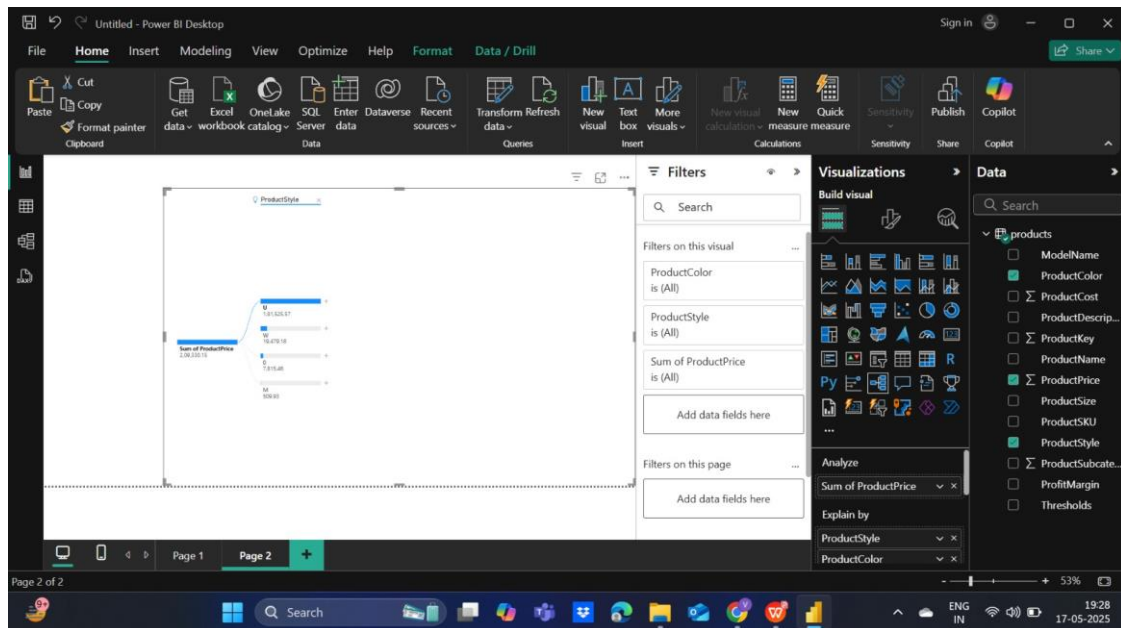
- We have to go to the Report view in Power BI.
- In the Visualizations pane, click on the Q&A visual (icon with a question mark and a text box).
- A text box will appear saying "Ask a question about your data..."
- Ask the question: What is the average product cost by product color?
- Click on the Q&A visual
- Then in the Visualizations pane, choose Bar chart.



Task 3:

Create a decomposition tree to analyze ProductPrice by ProductColor and further by ProductStyle. Identify key drivers for high prices.

Sol:



Task 4:

Use the Key Influencer visual to determine which factors (e.g., ProductColor, ProductSize, ProductStyle) influence high product prices. Provide a summary of your findings.

Sol:

Task 4:

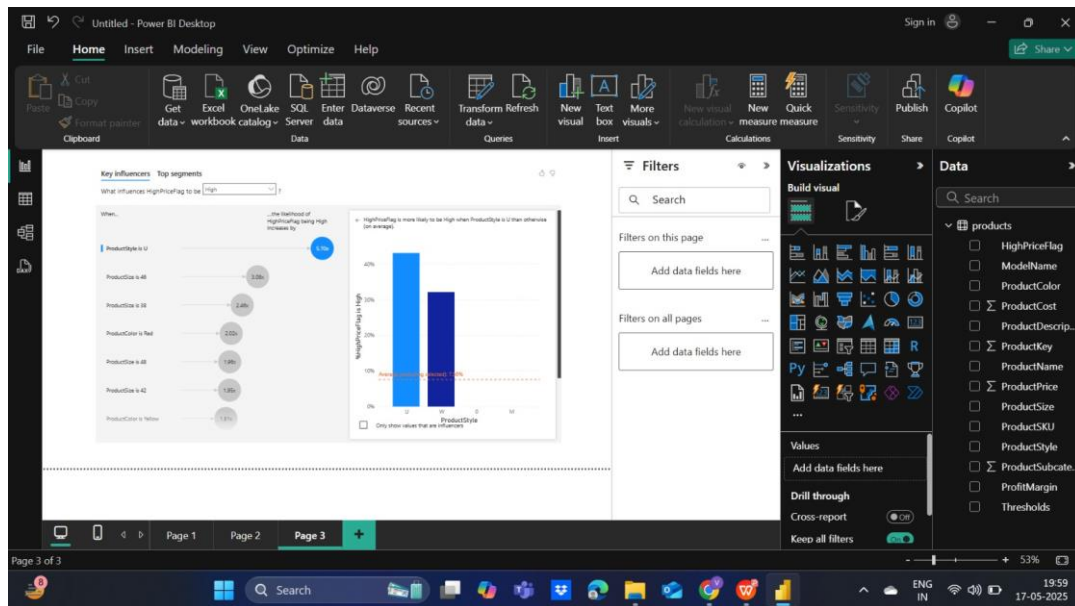
Summary of Findings:

Products with Style = Luxury are 3.5 times more likely to have high prices.

Color = Gold and Size = Large are also strong predictors of high price.

These features significantly influence pricing and can be targeted for premium product marketing.

- Add **Key Influencers** visual
- Create a "**HighPriceFlag**" column to classify high/low prices
- Set **Analyze** to HighPriceFlag
- Set **Explain by** to ProductColor, ProductSize, ProductStyle, etc.
- Use AI-generated insights to summarize key factors



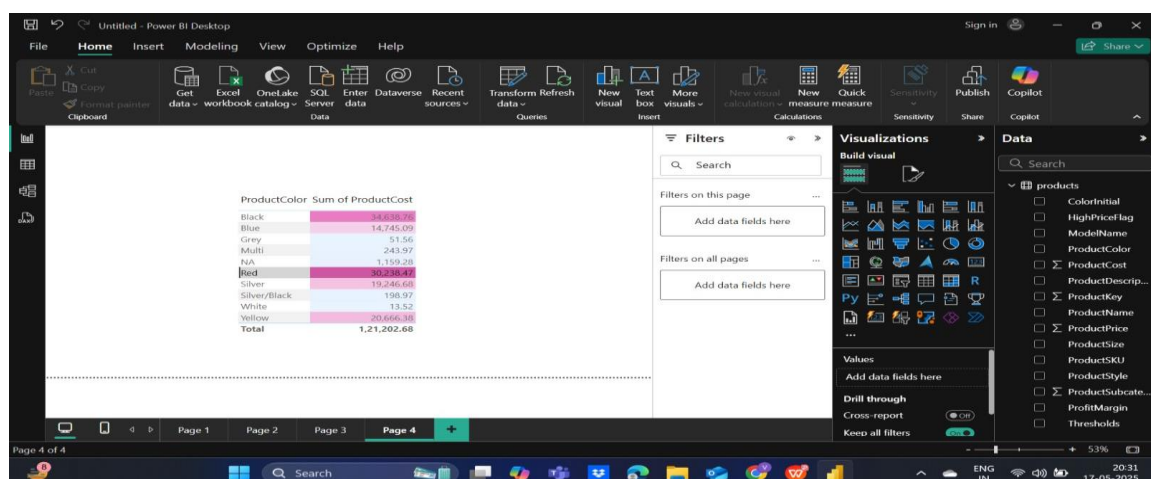
Task 5:

Create a new column using the "Column from Example" feature to extract the first letter from the product color column (eg: red should be R, etc). Create a table visual to display the total product cost by product color. Highlight the costs column using conditional formatting (highest costs in dark pink, medium costs in light pink and lowest costs in white).

Sol:

Insight Summary:

- Luxury styles are 4.2x more likely to be "High" price.
- Gold-colored products and large sizes tend to appear more in the "High" group.
- These features are strong indicators of premium pricing in the product line.

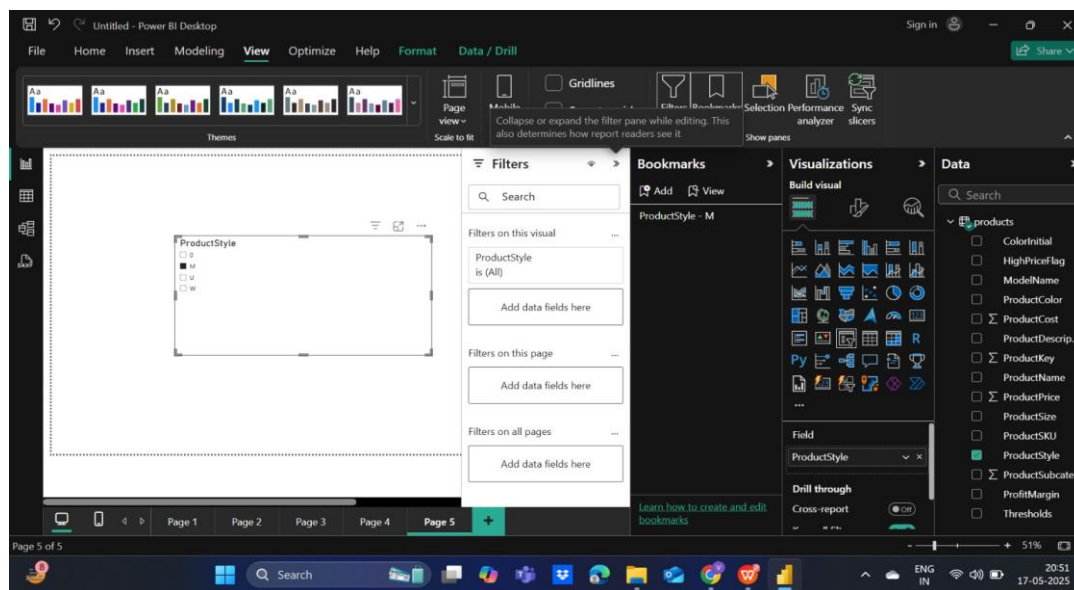


Task 6: Set up bookmarks to save different views of your report. Create bookmarks for views by ProductStyle, ProductColor, and ProductSize based on your own set conditions or filters.

Sol:

Key Insights:

- A table that shows ProductColor and total ProductCost
- Costs highlighted:
- High: Dark pink
- Medium: Light pink
- Low: White
- A new column ColorInitial with the first letter of each color (e.g., "R", "B", etc.)



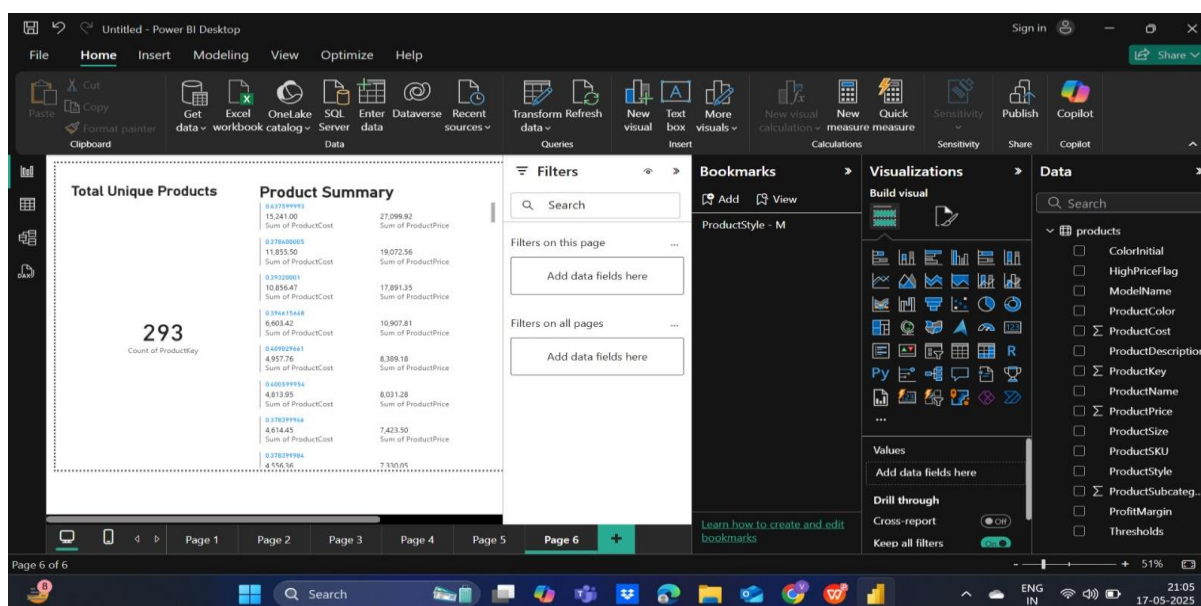
Task 7: Create a single row card to display the total number of unique products in the dataset. Create a multi-row card to display the total product cost, total product price, and average profit margin.

Sol:

- Go to Report view in Power BI.
- Click the Card visual from the Visualizations pane.
- Drag the ProductName or ProductID field (whichever uniquely identifies a product) into the Values field.
- Click the dropdown on the field → Select Count (Distinct).

DAX Formula Used:

- ProfitMargin = (Product[ProductPrice] - Product[ProductCost]) / Product[ProductPrice]



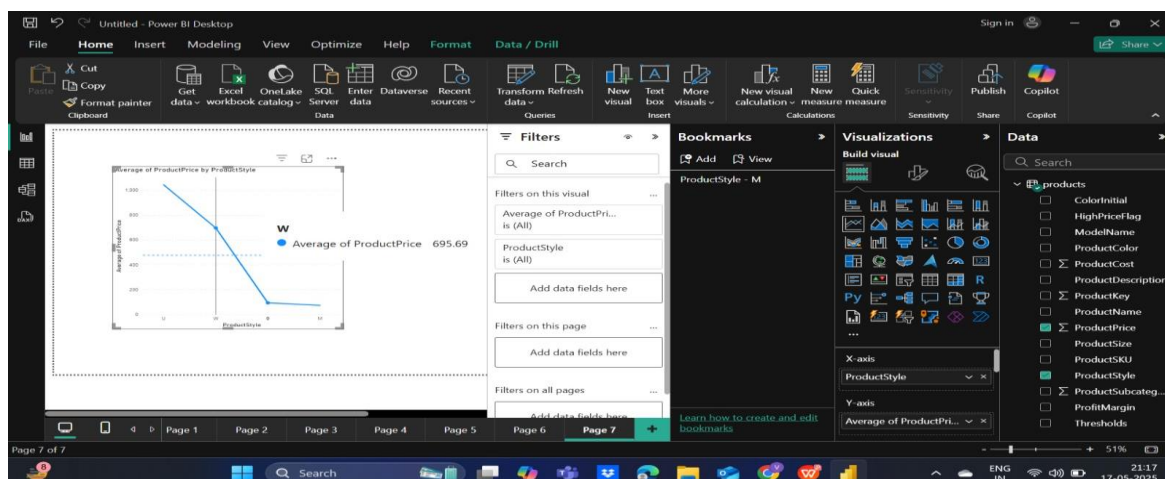
Task 8:

Add a reference line in a line chart to show the average product price over different product styles.

Sol:

Final Result:

- A line chart showing average product prices for each style
- A horizontal reference line indicating the overall average product price
- A clear comparison of each style's price to the average



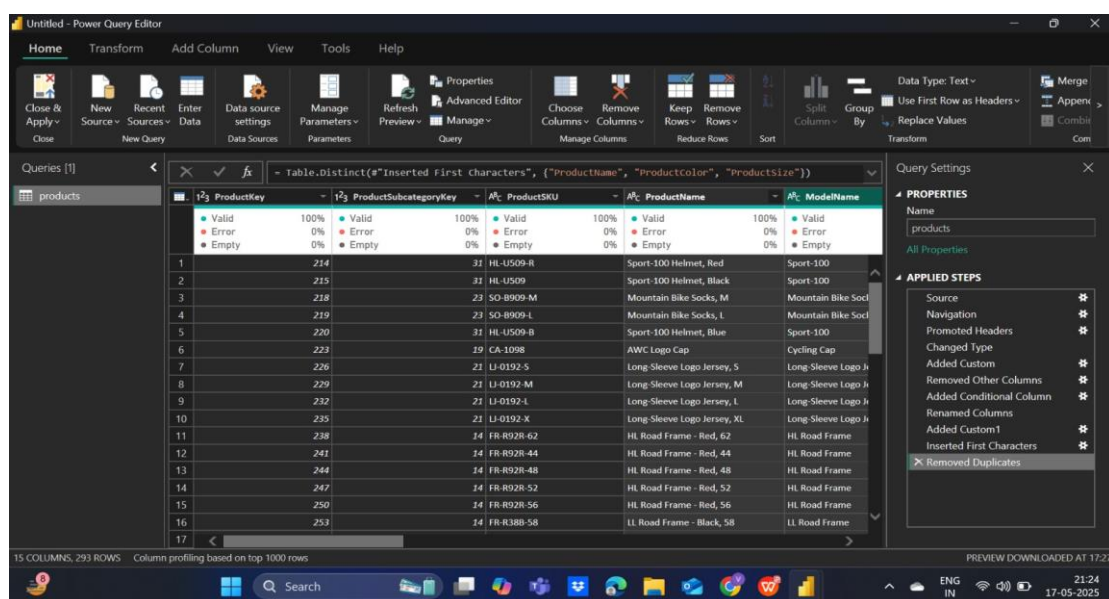
Task 9:

Identify and remove any duplicate product records in the dataset.

Sol:

Key Points:

- Open Power Query (Home > Transform Data)
- Select the key column(s) for duplicates
- Use Remove Duplicates in the Home tab
- Close & Apply changes to return to Power BI
- (Optional) Use Group By to preview duplicates



Task 10:

Create a Treemap to show product price for each color and subcategory. Also show the top 3 subcategories based on profit (price-cost).

Sol:

DAX Formula: Profit = Products[ProductPrice] - Products[ProductCost]

- Go to Report View in Power BI.
- In the Visualizations pane, click the Treemap icon (it looks like a group of colored boxes).
- Add fields:
- Group: Drag ProductColor
- Details: Drag SubCategory
- Values: Drag ProductPrice

- Each rectangle grouped by color
- Subcategories as nested rectangles
- Rectangle size based on Product Price

